

# DA-MambaIR :

## Degradation-Aware State Space Models for Unified Image Restoration

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# Content

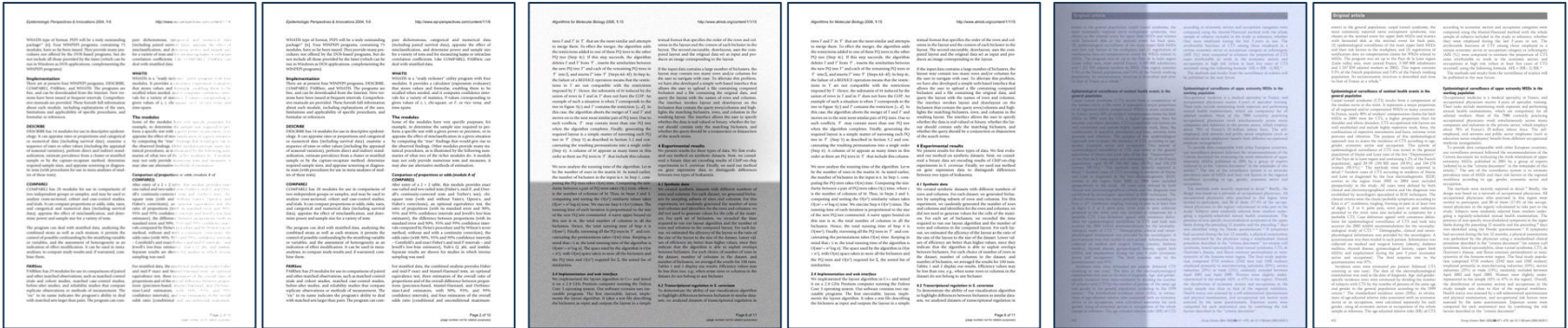
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- Introduction
- Proposed methods
- Experimental Results
- Conclusion

# Introduction

## Motivation : The Challenge of Real-world Restoration

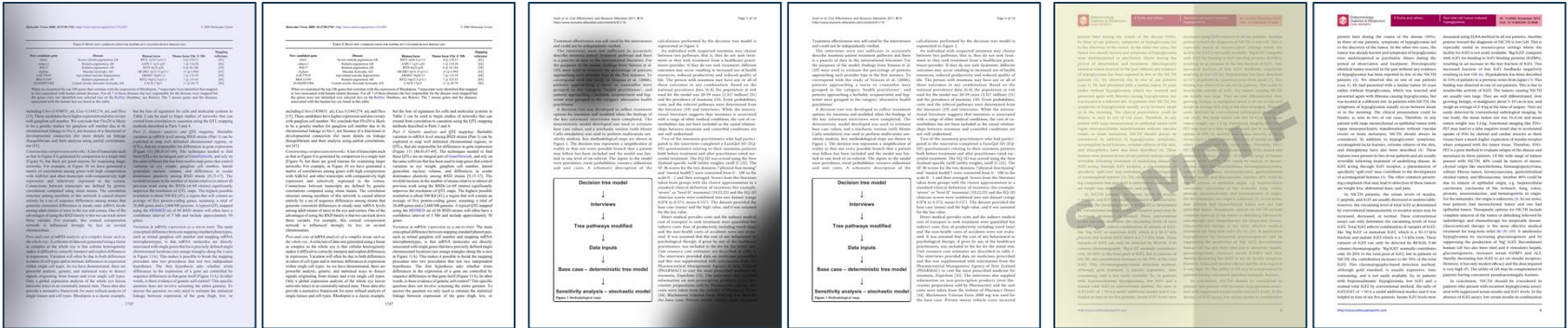
- While researching “Document Image Restoration”, we should address **mixtures** of Blur, Noise, Shadow, Folding **simultaneously**



Fold + Noise + Blur

Shadow + Noise + Fold

Color + Noise + Blur + Shadow + Fold



Color + Shadow

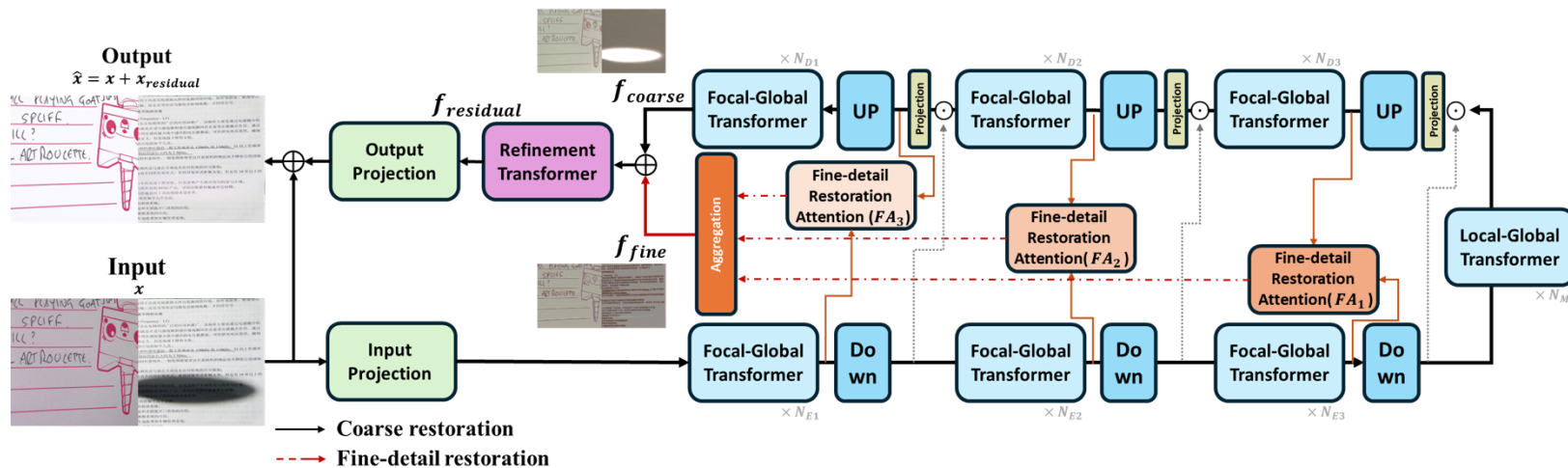
Shadow (Diverse Intensity)

Color + Fold + Watermark

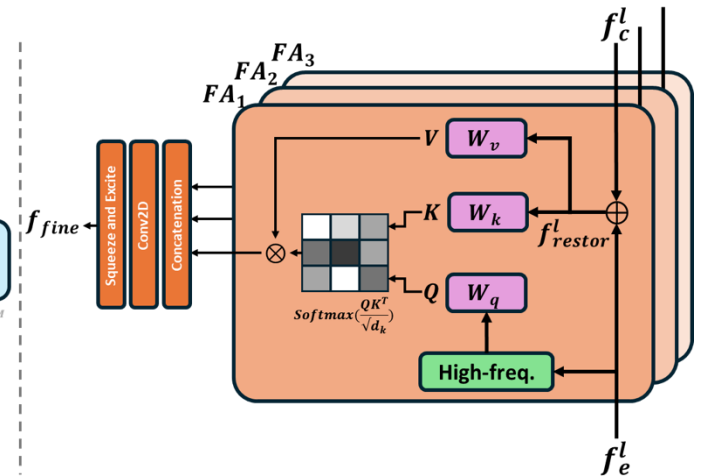
# Introduction

## Motivation : The Challenge of Real-world Restoration -> The Transformer Bottleneck

- Transformers are powerful but computationally prohibitive for high-resolution tasks
- Transformer-based models suffer from Quadratic Complexity due to (self-)attention mechanisms
- Even with an **RTX 4090 (24GB VRAM)**, memory is fully saturated processing just **one high-resolution image**



(a) Overview of Unified Document Image Restoration framework

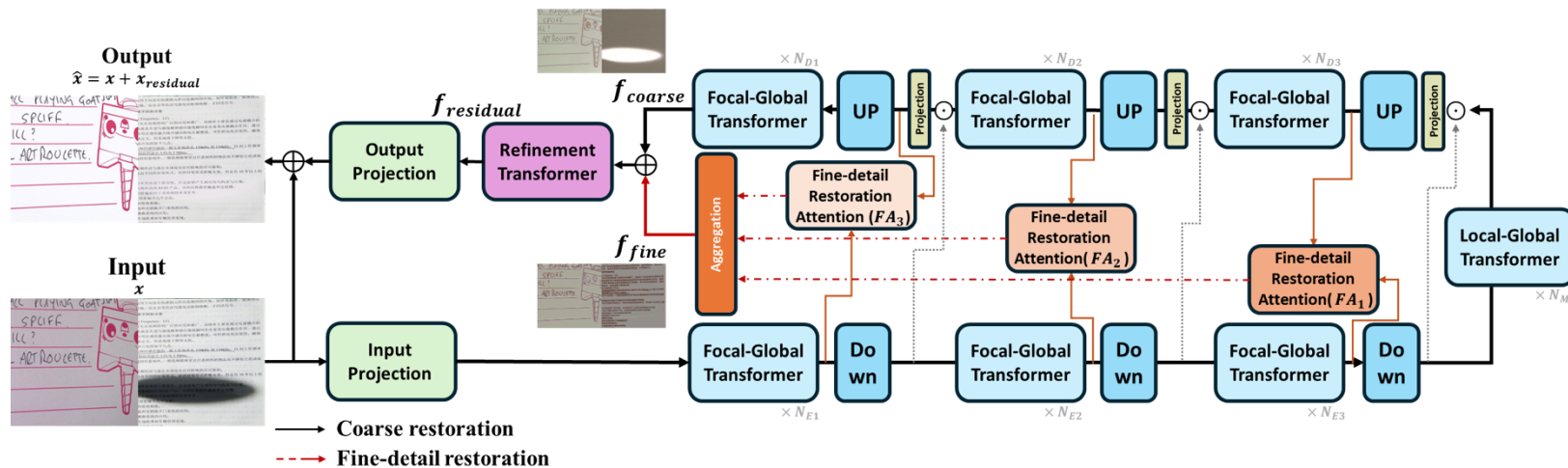


(b) Fine-detail Restoration Flow

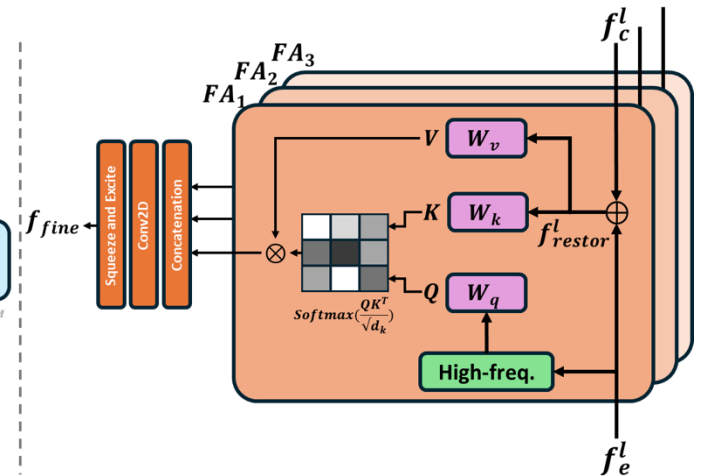
# Introduction

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(a) Overview of Unified Document Image Restoration framework



(b) Fine-detail Restoration Flow

➔ To tackle **more complex restoration, video restoration, and 4D scenes**, we need a **sustainable and scalable architecture**

# Introduction

Motivation : ... -> ... ->  mamba

## 1. Why use Mamba<sub>[1]</sub>?

Transformer

[Self-Attention]

- Strength: Long-range dependency
- Limitation: Quadratic computational complexity → Limits scalability

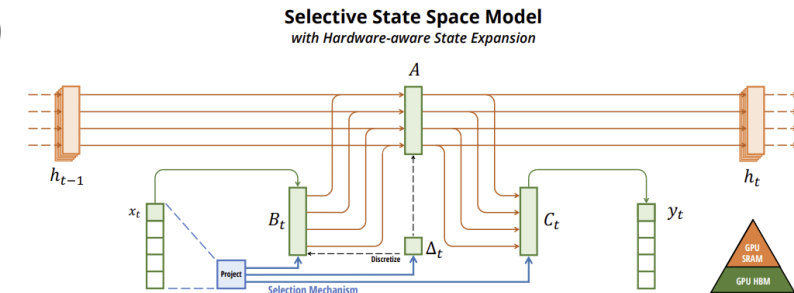


Address computational bottleneck

Mamba

[Selective-Scan Mechanism]

- Strength
  - ✓ Content-dependent feature selection with **linear complexity**
  - ✓ **Effective global receptive field** (good promise in IR)
- Limitation: Limited suitability for unified IR



[1] Gu, Albert, and Tri Dao. "Mamba: Linear-time sequence modeling with selective state spaces." *First conference on language modeling*. 2024.

# Introduction

Motivation : ... -> ... ->  **mamba** for Restoration

## 2. Why an All-in-One Approach Is Needed in Mamba-based Models

### Transformer

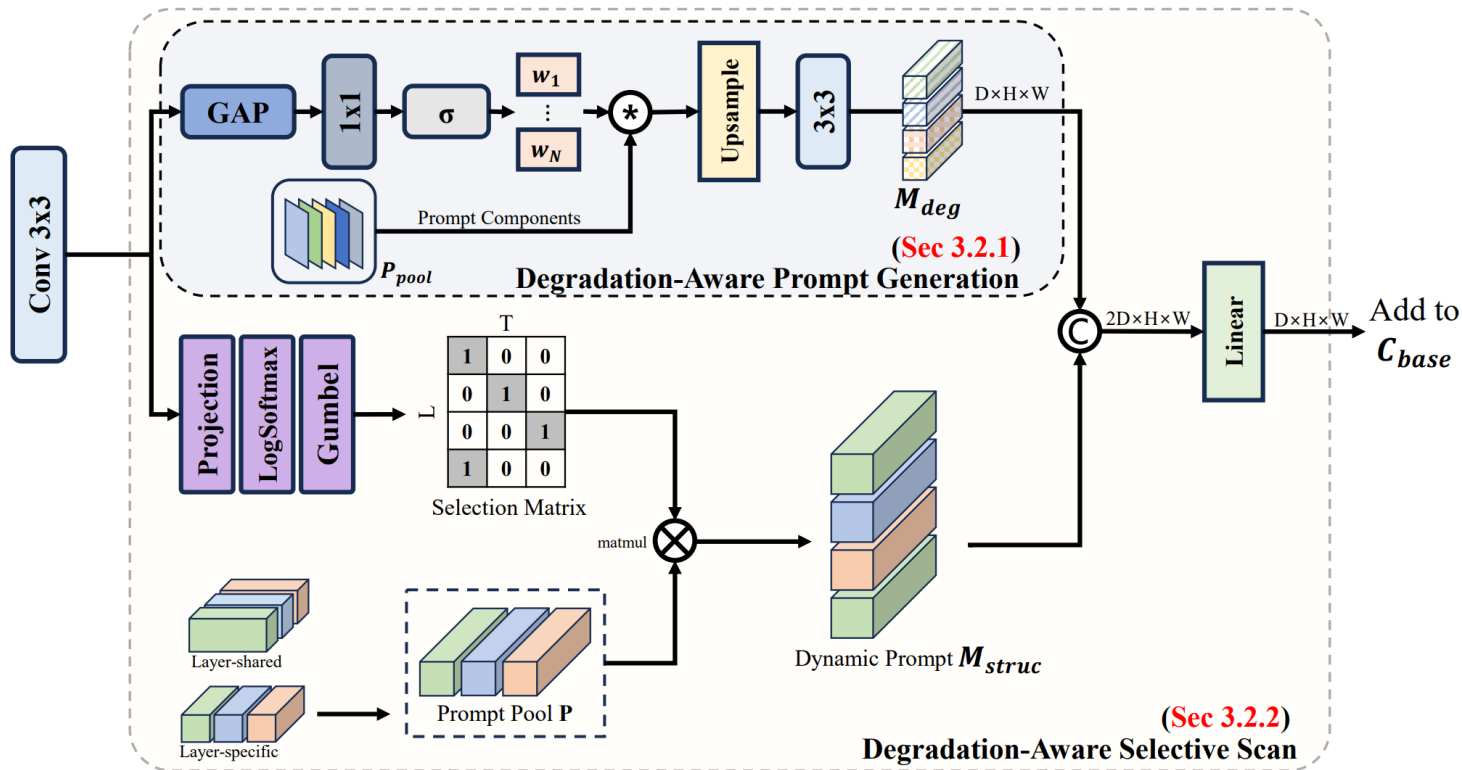
- Successfully employed task-aware priors to handle multiple degradations

### Mamba

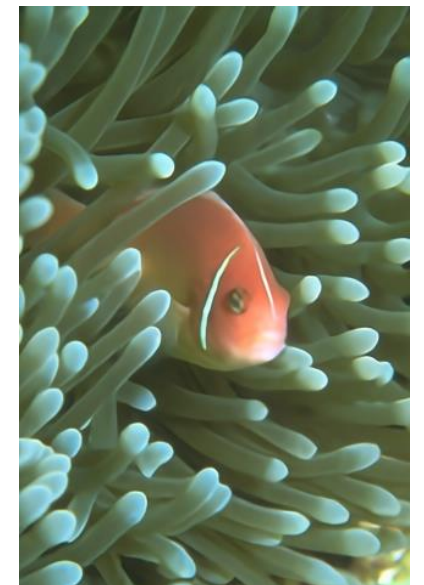
- **Selective-Scan Mechanism**
  - ✓ **Content-dependent Selection:** Unlike CNNs with static weights, Mamba dynamically adjusts parameters based on the input content.
  - ✓ **Adaptive Filtering:** This allows the model to flexibly distinguish and process different degradation types (noise, blur, rain) within a single unified framework.
- **Scalability for Composite Tasks**
  - ✓ **Linear Complexity:** Enables handling complex, composite degradation tasks efficiently without the computational bottleneck of Transformers.

# Introduction

## Goal : Degradation-aware Mamba-based Image Restoration



MambaIRv2[2]



Ours

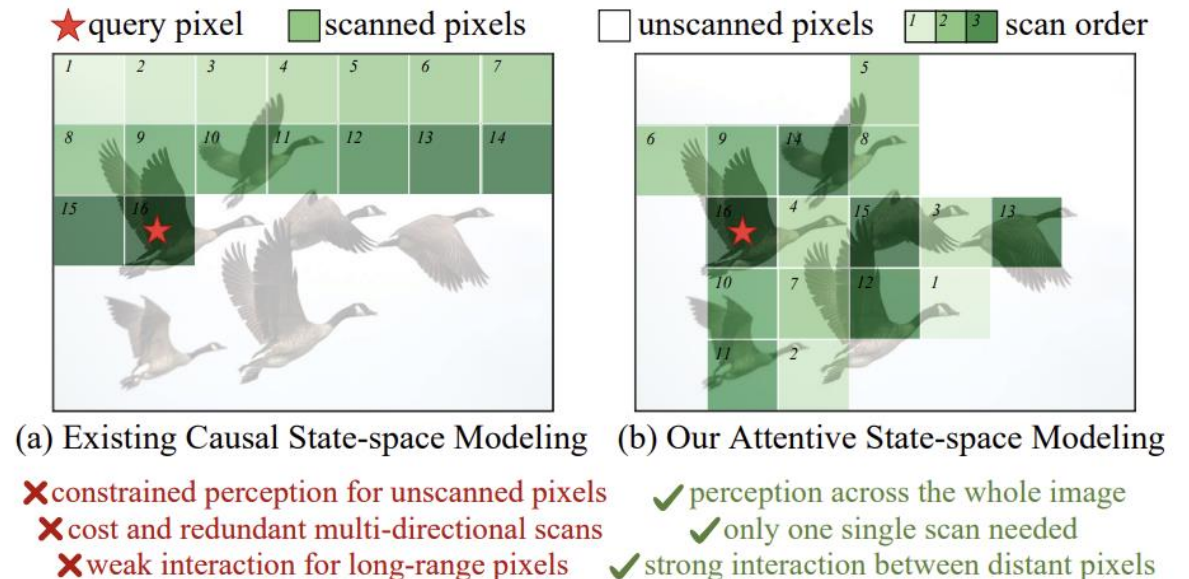
HR + Derain

# Related works

## “Redefine intrinsic state transition of MambaIRv2”

### MambaIRv2

- Pixel-wise Semantic Grouping
- Strength: Captures **structural dependencies** at the pixel level
- Limitation: Operates in a **degradation-agnostic manner**
  - Similar orientations or patterns may be confused as the same structure (e.g., rain streaks  $\approx$  vertical edges)



MambaIRv2 [2]

# Related works

## “Redefine intrinsic state transition of MambaRv2”

### MambaRv2

- Pixel-wise Semantic Grouping
- Strength: Captures structural dependencies at the pixel level
- Limitation: Operates in a degradation-agnostic manner
  - Similar orientations or patterns may be confused as the same structure (e.g., rain streaks  $\approx$  vertical edges)
- **Our Proposal: “Learnable Prompt Variables”**
  - Directly modulate the parameters of the Selective Scan operation

### PromptIR<sup>[3]</sup>

- Prompt at decoder input  $\rightarrow$  Indirect control
- **Our Proposal: Prompt inside SSM core**  $\rightarrow$  Degradation-aware semantic grouping

✓ **Key Idea: Structural patterns  $\oplus$  Characteristics of degradations**  $\rightarrow$  Reconfigure Selective Scan mechanism itself

[2] Guo, Hang, et al. "Mambairv2: Attentive state space restoration." Proceedings of the Computer Vision and Pattern Recognition Conference. 2025.

[3] Potlapalli, Vaishnav et al. "PromptIR: Prompting for All-in-One Image Restoration." Neural Information Processing Systems (2023).

# Proposed methods

## Our Framework

- ‘SSM linear efficiency’ + ‘Prompt learning adaptability’
- **Unified Image Restoration**
  - A single, efficient architecture for multiple degradation types
- **Degradation-Aware Prompt Integration**
  - Dynamically modulates intrinsic parameters of Selective Scan
  - Identifies degradation patterns → adapts semantic grouping accordingly

**SSM-based IR vs DA-MambaIR (Ours)**

| SSM-based IR      | DA-MambaIR              |
|-------------------|-------------------------|
| Task-agnostic     | Degradation-aware       |
| Structure-focused | Structure + degradation |
| Static grouping   | Adaptive grouping       |

# Proposed methods

## Preliminaries: State Space Models

### State Space Models

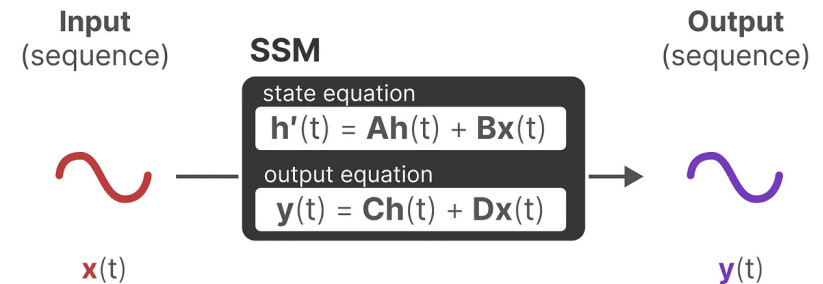
- Linear Time-Invariant SSM
  - Static Transition parameters
  - State Equation:  $h_k = \bar{A}h_{k-1} + \bar{B}x_k$
  - Output Equation:  $y_k = Ch_k$

### DA-MambaIR

- Leverage input-dependence → Introduce SSM degradation-dependent

### Mamba: Selective Scan

- Input-dependent B, C,  $\Delta$
- Context-aware propagation and forgetting



# Proposed methods

## (1) Degradation-Aware Prompt Generation

### 1. Global Degradation Descriptor

$$v = \text{GAP}(X) \in \mathbb{R}^c \quad X : \text{Input Image(Feature)}, \text{GAP} : \text{Global Average Pooling}$$

- Captures global degradation characteristics of the input images

### 2. Prompt Selection Weights

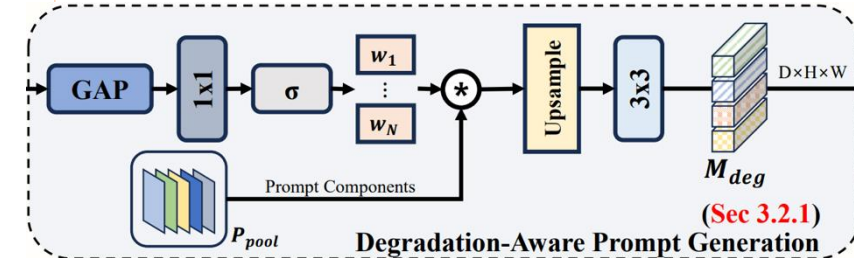
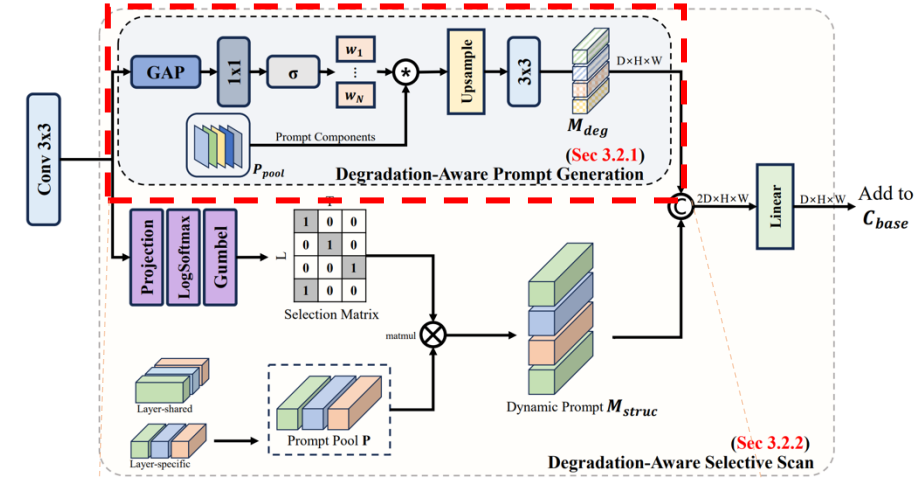
$$w = \text{Softmax}(\text{MLP}(v))$$

- Identify the dominant degradation type by selecting relevant prompts from the prompt pool

### 3. Pixel-wise Degradation Prompt Map

$$\mathbf{M}_{\text{deg}} = \text{Upsample} \left( \sum_{k=1}^N w_k \mathbf{P}_k \right) \in \mathbb{R}^{H \times W \times C}$$

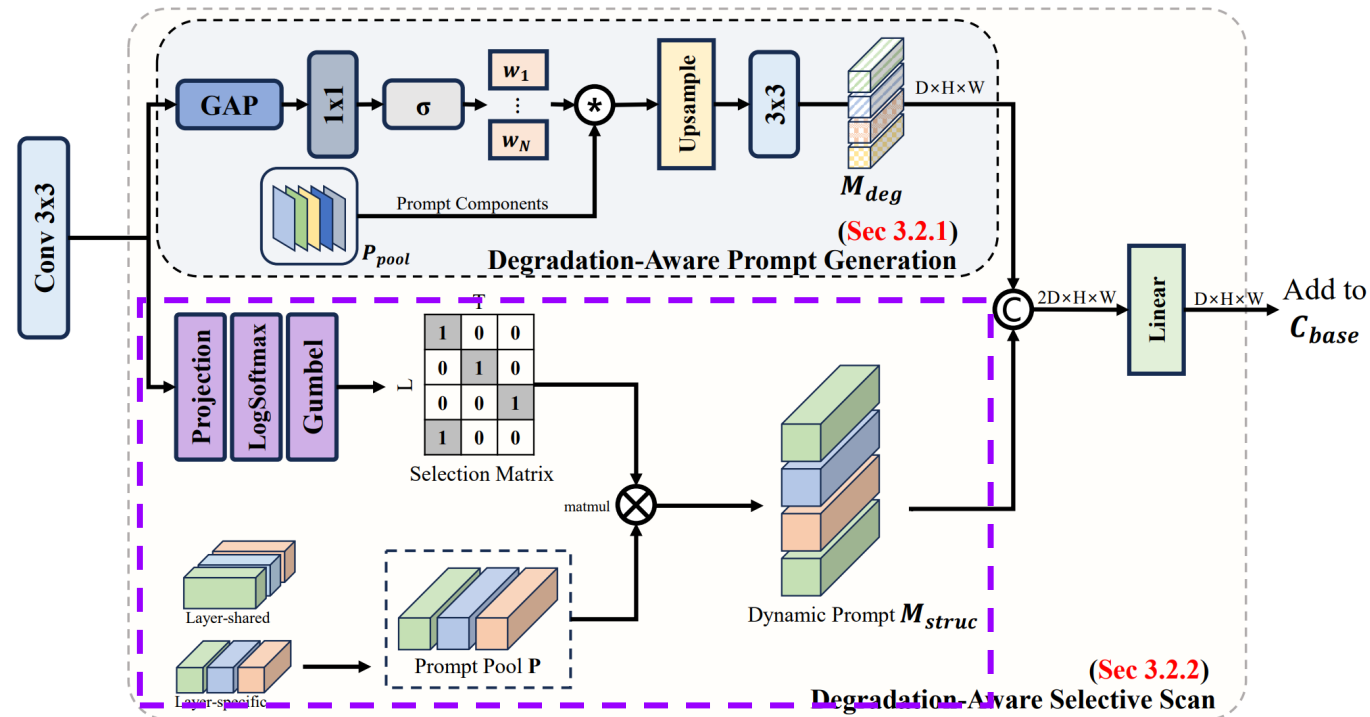
- Provides **spatially dense degradation guidance**, encoding where and how strongly degradations occur



# Proposed methods

## (2) Degradation-Aware Selective Scan

- C (Output Projection Parameter): Criterion for pixel-level semantic grouping
- **Our Proposal**
  - **Structural prompt (MambaRv2)  $\oplus$  Degradation Information  $\rightarrow$  Semantic Grouping Modulation**
    - Directly inject degradation cues into semantic grouping



# Proposed methods

## (2) Degradation-Aware Selective Scan

### 1. Prompt Fusion

$$\mathbf{Z} = \text{Linear}_{\text{fuse}} \left( \begin{array}{c} \text{(Concatenation)} \\ [\mathbf{M}_{\text{struc}}; \mathbf{M}_{\text{deg}}] \end{array} \right)$$

(Structural Prompt from MambaRv2) (Degradation-aware Prompt)

- Structural context  $\oplus$  Degradation information

### 2. Base Projection

$$\mathbf{C}_{\text{base}} = \text{Linear}_C(\mathbf{x}_t)$$

- Content-driven semantic grouping derived from input  $x_t$

### 3. Degradation-Aware Projection

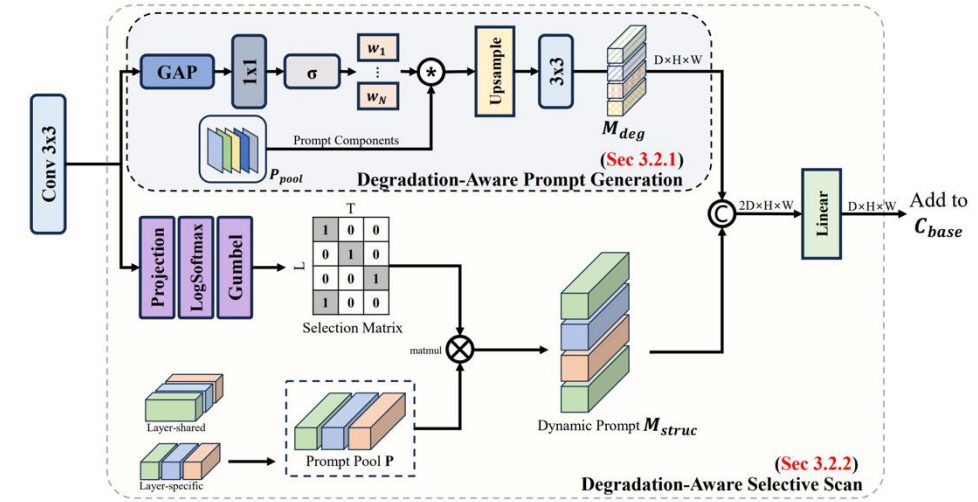
$$\hat{\mathbf{C}}_t = \mathbf{C}_{\text{base}} + \mathbf{Z}$$

- Recalibrate semantic grouping by injecting degradation-aware modulation into the projection parameter

### 4. Output Generation

$$\mathbf{y}_t = \hat{\mathbf{C}}_t \mathbf{h}_t$$

- SSM output with degradation-aware semantic grouping
- Filter out artifacts while preserving underlying image details



# Proposed methods

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## Training

- Training Objective
  - Mean Squared Error(MSE) loss

$$\mathcal{L} = \left\| \hat{\mathbf{I}} - \mathbf{I}_{GT} \right\|_2^2$$

# Experiments

## Datasets

- 3 representative benchmarks
  1. **Rain100L**: Deraining
    - Rain streak removal dataset
    - Rainy images with diverse streak directions and intensities
  2. **SOTS(RESIDE)**: Dehazing
    - Outdoor subset
    - Evaluate visibility and color restoration under haze
  3. **Urban100**: Denoising
    - Gaussian noise added with  $\sigma = 15$
    - Focuses on noise suppression while preserving fine details
- **Dual-task training setup: Image Restoration + Super Resolution**
  - e.g., Joint SR-Deraining, Joint SR-Dehazing, and Joint SR-Denoising

## Evaluation metrics

- **PSNR**: Pixel-level fidelity to GT
- **SSIM**: Structural consistency and perceptual quality

# Experiments

## Quantitative Results

| Task       | Dataset       | MambaIRv2 [7] |         | MambaIRv2*     |               | Ours           |               |
|------------|---------------|---------------|---------|----------------|---------------|----------------|---------------|
|            |               | PSNR(↑)       | SSIM(↑) | PSNR(↑)        | SSIM(↑)       | PSNR(↑)        | SSIM(↑)       |
| De-raining | Rain100L [11] | 24.3390       | 0.6664  | <b>27.2777</b> | <b>0.7464</b> | 27.1939        | 0.7460        |
| De-hazing  | SOTS [12]     | 16.9130       | 0.6739  | 25.9428        | 0.7682        | <b>26.0705</b> | <b>0.7702</b> |
| Denoising  | Urban100 [13] | 22.1783       | 0.5165  | 25.2938        | <b>0.7479</b> | <b>25.3047</b> | 0.7465        |

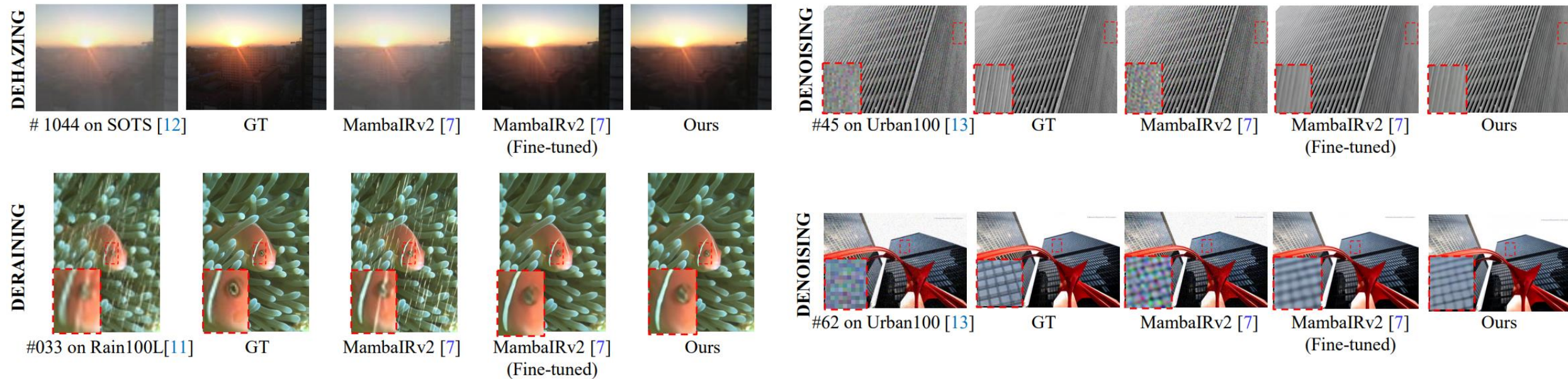
\* Denotes fine-tuned on composite degradation datasets  
(e.g., LR->HR, De-rain)

- De-raining (Rain100L)
  - Slightly lower PSNR and SSIM, but difference is negligible
- De-hazing (SOTS)
  - Best PSNR / SSIM (26.07dB / 0.7702)
  - Outperforms single-task expert
- Denoising (Urban100)
  - Best PSNR (25.30 dB)
  - Slightly lower SSIM, but difference is marginal

⇒ By dynamically modulating state-space parameters,  
DA-MambaIR adapts to diverse degradations without negative transfer

# Experiments

## Qualitative Results



**Figure 2.** Qualitative comparison on image restoration tasks, including dehazing, deraining, and denoising.

# Conclusion

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## Summary

### Overview

- Unified image restoration framework
- Single model for multiple degradation types

### Key Idea

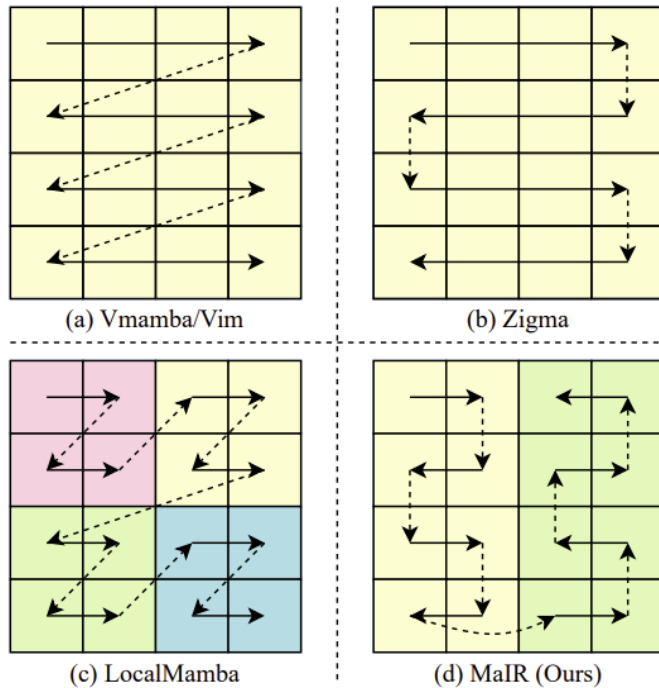
- **Degradation-aware prompts injected into the State Space Model**
- Direct modulation of projection parameter  $\hat{\mathbf{C}}_t$

⇒ **Dynamic state space modulation for scalable and robust low-level vision systems**

# Conclusion

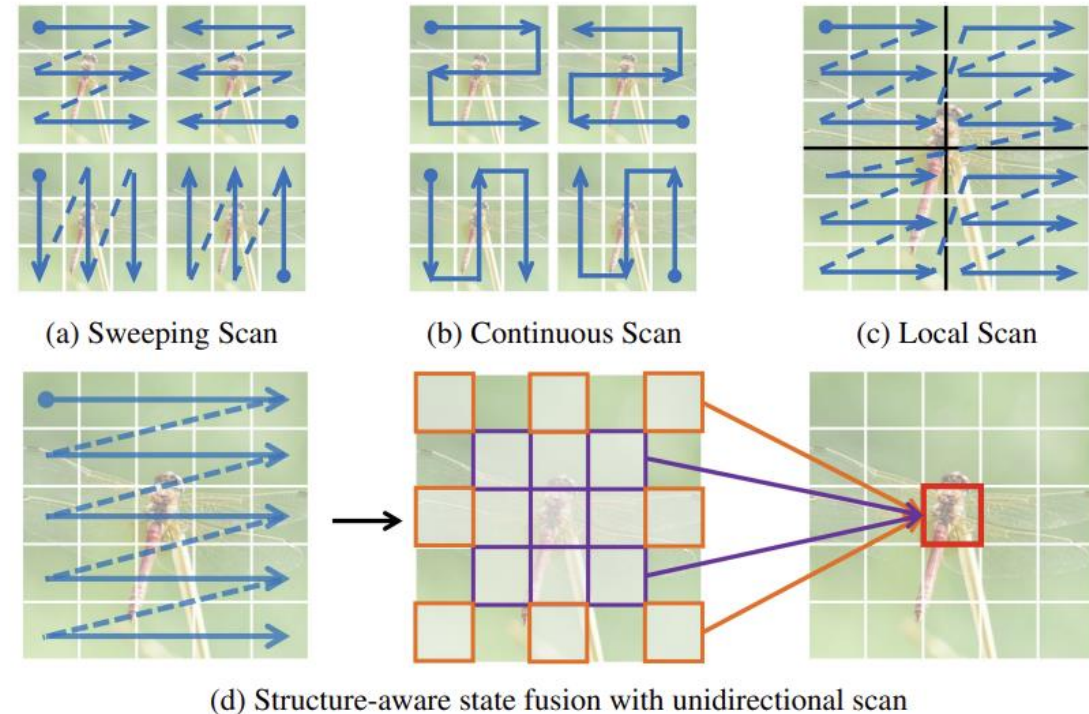
## Future works

- Explore the degradation-aware SSM strategy (w/ contrative learning)



**MAIR** [4]

- Expand to Video restoration tasks



**Spatial-Mamba** [5]

[4] Li, Boyun, et al. "Mair: A locality-and continuity-preserving mamba for image restoration." Proceedings of the Computer Vision and Pattern Recognition Conference. 2025..

[5] Xiao, Chaodong, et al. "Spatial-mamba: Effective visual state space models via structure-aware state fusion." ICLR 2025.

# Thank you

## Q&A